Homework 2

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Question 6

(a) $\forall x\, P(x) \vdash \forall y\, P(y)$

1.	$\forall x\, P(x)$	premise
2.	y_0	fresh variable
3.	$P(y_0)$	$\forall$e, 1
4.	$\forall y\, P(y)$	$\forall$i, 2–3

(b) $\forall x\, (P(x) \to Q(x)) \vdash (\forall x\, \neg Q(x)) \to (\forall x\, \neg P(x))$

1.	$\forall x\, (P(x) \to Q(x))$	premise
2.	$\forall x\, \neg Q(x)$	assumption
3.	x_0	fresh variable
4.	$P(x_0) \to Q(x_0)$	$\forall$e, 1
5.	$\neg Q(x_0)$	$\forall$e, 2
6.	$P(x_0)$	assumption
7.	$Q(x_0)$	$\to$e, 4, 6
8.	$\perp$	$\neg$e, 5, 7
9.	$\neg P(x_0)$	$\neg$i, 6–8
10.	$\forall x\, \neg P(x)$	$\forall$i, 3–9
11.	$(\forall x\, \neg Q(x)) \to (\forall x\, \neg P(x))$	$\to$i, 2–10

(c) $\forall x\,(P(x) \to \neg Q(x)) \vdash \neg(\exists x\,(P(x) \wedge Q(x)))$

1.	$\forall x\,(P(x) \to \neg Q(x))$	premise
2.	$\exists x\,(P(x) \wedge Q(x))$	assumption
3.	x_0	fresh variable
4.	$P(x_0) \wedge Q(x_0)$	assumption
5.	$P(x_0) \to \neg Q(x_0)$	$\forall$e, 1
6.	$P(x_0)$	$\wedge$e$_1$, 4
7.	$Q(x_0)$	$\wedge$e$_2$, 4
8.	$\neg Q(x_0)$	$\to$e, 5, 6
9.	$\bot$	$\neg$e, 7, 8
10.	$\bot$	$\exists$e, 2, 3–9
11.	$\neg(\exists x\,(P(x) \wedge Q(x)))$	$\neg$i, 2–10

Question 9

(b) $S \to \exists x\, Q(x) \vdash \exists x\,(S \to Q(x))$

1.	$S \to \exists x\, Q(x)$	premise
2.	$S \vee \neg S$	LEM
3.	S	assumption
4.	$\exists x\, Q(x)$	$\to$e, 1, 3
5.	x_0	fresh variable
6.	$Q(x_0)$	assumption
7.	S	assumption
8.	$Q(x_0)$	
9.	$S \to Q(x_0)$	$\to$i, 7–8
10.	$\exists x\,(S \to Q(x))$	$\exists$i, 9
11.	$\exists x\,(S \to Q(x))$	$\exists$e, 4, 5–10
12.	$\neg S$	assumption
13.	x_0	fresh variable
14.	S	assumption
15.	$\bot$	$\neg$e, 12, 14
16.	$Q(x_0)$	$\bot$e, 15
17.	$S \to Q(x_0)$	$\to$i, 14–16
18.	$\exists x\,(S \to Q(x))$	$\exists$i, 17
19.	$\exists x\,(S \to Q(x))$	
20.	$\exists x\,(S \to Q(x))$	$\vee$e, 2, 3–11, 12–19

(e) $\forall x\,(P(x) \lor Q(x)) \vdash \forall x\,P(x) \lor \exists x\,Q(x)$

1.	$\forall x\,(P(x) \lor Q(x))$	premise
2.	$\forall x\,P(x) \lor \neg\forall x\,P(x)$	LEM
3.	$\forall x\,P(x)$	assumption
4.	$\forall x\,P(x) \lor \exists x\,Q(x)$	$\lor$i$_1$, 3
5.	$\neg\forall x\,P(x)$	assumption
6.	$\neg\forall x\,P(x) \to \exists x\,\neg P(x)$	quantifier equivalence
7.	$\exists x\,\neg P(x)$	$\to$e, 5, 6
8.	x_0	fresh variable
9.	$\neg P(x_0)$	assumption
10.	$P(x_0) \lor Q(x_0)$	$\forall$e, 1
11.	$P(x_0)$	assumption
12.	$\bot$	$\neg$e, 9, 11
13.	$Q(x_0)$	$\bot$e, 12
14.	$Q(x_0)$	assumption
15.	$Q(x_0)$	
16.	$Q(x_0)$	$\lor$e, 10, 11–13, 14–15
17.	$\exists x\,Q(x)$	$\exists$i, 16
18.	$\forall x\,P(x) \lor \exists x\,Q(x)$	$\lor$i$_2$, 17
19.	$\forall x\,P(x) \lor \exists x\,Q(x)$	$\exists$e, 7, 8–18
20.	$\forall x\,P(x) \lor \exists x\,Q(x)$	$\lor$e, 2, 3–4, 5–19

(i) $\exists x\,(\neg P(x) \land \neg Q(x)) \vdash \exists x\,(\neg(P(x) \land Q(x)))$

1.	$\exists x\,(\neg P(x) \land \neg Q(x))$	premise
2.	x_0	fresh variable
3.	$\neg P(x_0) \land \neg Q(x_0)$	assumption
4.	$\neg P(x_0)$	$\land$e$_1$, 3
5.	$P(x_0) \land Q(x_0)$	assumption
6.	$P(x_0)$	$\land$e$_1$, 5
7.	$\bot$	$\neg$e, 4, 6
8.	$\neg(P(x_0) \land Q(x_0))$	$\neg$i, 5–7
9.	$\exists x\,(\neg(P(x) \land Q(x)))$	$\exists$i, 8
10.	$\exists x\,(\neg(P(x) \land Q(x)))$	$\exists$e, 1, 2–9

Question 11

(b) $P(b),\ \forall x \forall y\,(P(x) \wedge P(y) \to x = y) \vdash \forall x\,(P(x) \leftrightarrow x = b)$

1.	$P(b)$	premise
2.	$\forall x \forall y\,(P(x) \wedge P(y) \to x = y)$	premise
3.	x_0	fresh variable
4.	$\forall y\,(P(x_0) \wedge P(y) \to x_0 = y)$	$\forall$e, 2
5.	$P(x_0) \wedge P(b) \to x_0 = b$	$\forall$e, 4
6.	$P(x_0)$	assumption
7.	$P(x_0) \wedge P(b)$	$\wedge$i, 6, 1
8.	$x_0 = b$	$\to$e, 5, 7
9.	$P(x_0) \to x_0 = b$	$\to$i, 6–8
10.	$x_0 = b$	assumption
11.	$P(x_0)$	=e, 10, 1
12.	$x_0 = b \to P(x_0)$	$\to$i, 10–11
13.	$(P(x_0) \to x_0 = b) \wedge (x_0 = b \to P(x_0))$	$\wedge$i, 9, 12
14.	$\forall x\,(P(x) \leftrightarrow x = b)$	$\forall$i, 3–13

(d) $\forall x\,(P(x) \leftrightarrow x = b) \vdash P(b) \wedge \forall x \forall y\,(P(x) \wedge P(y) \rightarrow x = y)$

1.	$\forall x\,(P(x) \leftrightarrow x = b)$	premise
2.	$(P(b) \rightarrow b = b) \wedge (b = b \rightarrow P(b))$	$\forall$e, 1
3.	$b = b \rightarrow P(b)$	$\wedge \mathrm{e}_2$, 2
4.	$b = b$	=i
5.	$P(b)$	$\rightarrow$e, 3, 4
6.	x_0	fresh variable
7.	y_0	fresh variable
8.	$P(x_0) \wedge P(y_0)$	assumption
9.	$P(x_0)$	$\wedge \mathrm{e}_1$, 8
10.	$P(y_0)$	$\wedge \mathrm{e}_2$, 8
11.	$(P(x_0) \rightarrow x_0 = b) \wedge (x_0 = b \rightarrow P(x_0))$	$\forall$e, 1
12.	$(P(y_0) \rightarrow y_0 = b) \wedge (y_0 = b \rightarrow P(y_0))$	$\forall$e, 1
13.	$P(x_0) \rightarrow x_0 = b$	$\wedge \mathrm{e}_1$, 11
14.	$P(y_0) \rightarrow y_0 = b$	$\wedge \mathrm{e}_1$, 12
15.	$x_0 = b$	$\rightarrow$e, 13, 9
16.	$y_0 = b$	$\rightarrow$e, 14, 10
17.	$x_0 = y_0$	=e, 15, 16
18.	$P(x_0) \wedge P(y_0) \rightarrow x_0 = y_0$	$\rightarrow$i, 8–17
19.	$\forall y\,(P(x_0) \wedge P(y) \rightarrow x_0 = y)$	$\forall$i, 7–18
20.	$\forall x \forall y\,(P(x) \wedge P(y) \rightarrow x = y)$	$\forall$i, 6–19
21.	$P(b) \wedge \forall x \forall y\,(P(x) \wedge P(y) \rightarrow x = y)$	$\wedge$i, 5, 20

Question 13

(e) $\forall x\,(P(x) \lor Q(x)),\ \exists x\,\neg Q(x),\ \forall x\,(R(x) \to \neg P(x)) \vdash \exists x\,\neg R(x)$

1.	$\forall x\,(P(x) \lor Q(x))$	premise
2.	$\exists x\,\neg Q(x)$	premise
3.	$\forall x\,(R(x) \to \neg P(x))$	premise
4.	x_0	fresh variable
5.	$\neg Q(x_0)$	assumption
6.	$P(x_0) \lor Q(x_0)$	$\forall$e, 1
7.	$P(x_0)$	assumption
8.	$P(x_0)$	
9.	$Q(x_0)$	assumption
10.	$\bot$	$\neg$e, 5, 9
11.	$P(x_0)$	$\bot$e, 10
12.	$P(x_0)$	$\lor$e, 6, 7–8, 9–11
13.	$R(x_0) \to \neg P(x_0)$	$\forall$e, 3
14.	$R(x_0)$	assumption
15.	$\neg P(x_0)$	$\to$e, 13, 14
16.	$\bot$	$\neg$e, 12, 15
17.	$\neg R(x_0)$	$\neg$i, 14–16
18.	$\exists x\,\neg R(x)$	$\exists$i, 17
19.	$\exists x\,\neg R(x)$	$\exists$e, 2, 4–18

(f) $\forall x\,(P(x) \to (Q(x) \lor R(x))),\ \neg\exists x\,(P(x) \land R(x)) \vdash \forall x\,(P(x) \to Q(x))$

1.	$\forall x\,(P(x) \to (Q(x) \lor R(x)))$	premise
2.	$\neg\exists x\,(P(x) \land R(x))$	premise
3.	x_0	fresh variable
4.	$P(x_0) \to (Q(x_0) \lor R(x_0))$	$\forall$e, 1
5.	$P(x_0)$	assumption
6.	$Q(x_0) \lor R(x_0)$	$\to$e, 4, 5
7.	$R(x_0)$	assumption
8.	$P(x_0) \land R(x_0)$	$\land$i, 5, 7
9.	$\exists x\,(P(x) \land R(x))$	$\exists$i, 8
10.	$\bot$	$\neg$e, 2, 9
11.	$Q(x_0)$	$\bot$e, 10
12.	$Q(x_0)$	assumption
13.	$Q(x_0)$	
14.	$Q(x_0)$	$\lor$e, 6, 7–11, 12–13
15.	$P(x_0) \to Q(x_0)$	$\to$i, 5–14
16.	$\forall x\,(P(x) \to Q(x))$	$\forall$i, 3–15

(g) $\exists x \exists y\,(S(x,y) \lor S(y,x)) \vdash \exists x \exists y\, S(x,y)$

1.	$\exists x \exists y\,(S(x,y) \lor S(y,x))$	premise
2.	x_0	fresh variable
3.	$\exists y\,(S(x_0,y) \lor S(y,x_0))$	assumption
4.	y_0	fresh variable
5.	$S(x_0,y_0) \lor S(y_0,x_0)$	assumption
6.	$S(x_0,y_0)$	assumption
7.	$\exists y\, S(x_0,y)$	$\exists$i, 6
8.	$\exists x \exists y\, S(x,y)$	$\exists$i, 7
9.	$S(y_0,x_0)$	assumption
10.	$\exists y\, S(y_0,y)$	$\exists$i, 9
11.	$\exists x \exists y\, S(x,y)$	$\exists$i, 10
12.	$\exists x \exists y\, S(x,y)$	$\lor$e, 5, 6–8, 9–11
13.	$\exists x \exists y\, S(x,y)$	$\exists$e, 3, 4–12
14.	$\exists x \exists y\, S(x,y)$	$\exists$e, 1, 2–13

(h) $\exists x\,(P(x) \wedge Q(x)),\ \forall y\,(P(y) \to R(y)) \vdash \exists x\,(R(x) \wedge Q(x))$

1.	$\exists x\,(P(x) \wedge Q(x))$	premise
2.	$\forall y\,(P(y) \to R(y))$	premise
3.	x_0	fresh variable
4.	$P(x_0) \wedge Q(x_0)$	assumption
5.	$P(x_0)$	$\wedge$e$_1$, 4
6.	$Q(x_0)$	$\wedge$e$_2$, 4
7.	$P(x_0) \to R(x_0)$	$\forall$e, 2
8.	$R(x_0)$	$\to$e, 5, 7
9.	$R(x_0) \wedge Q(x_0)$	$\wedge$i, 8, 6
10.	$\exists x\,(R(x) \wedge Q(x))$	$\exists$i, 9
11.	$\exists x\,(R(x) \wedge Q(x))$	$\exists$e, 1, 3–10

Question 14

Let $T(x)$ mean "x is a tax payer," $P(x)$ mean "x is a politician," and $H(x)$ mean "x is a philanthropist."

$(\exists x\, T(x) \to \forall x\, (P(x) \to T(x))),\ (\exists x\, H(x) \to \forall x\, (T(x) \to H(x))) \vdash (\exists x\, (T(x) \wedge H(x)) \to \forall x\, (P(x) \to H(x)$

1.	$\exists x\, T(x) \to \forall x\, (P(x) \to T(x))$	premise
2.	$\exists x\, H(x) \to \forall x\, (T(x) \to H(x))$	premise
3.	$\exists x\, (T(x) \wedge H(x))$	assumption
4.	x_0	fresh variable
5.	$T(x_0) \wedge H(x_0)$	assumption
6.	$T(x_0)$	$\wedge$e$_1$, 5
7.	$H(x_0)$	$\wedge$e$_2$, 5
8.	$\exists x\, T(x)$	$\exists$i, 6
9.	$\exists x\, H(x)$	$\exists$i, 7
10.	$\exists x\, T(x)$	$\exists$e, 3, 4–8
11.	x_1	fresh variable
12.	$T(x_1) \wedge H(x_1)$	assumption
13.	$T(x_1)$	$\wedge$e$_1$, 11
14.	$H(x_1)$	$\wedge$e$_2$, 11
15.	$\exists x\, T(x)$	$\exists$i, 12
16.	$\exists x\, H(x)$	$\exists$i, 13
17.	$\exists x\, H(x)$	$\exists$e, 3, 10–14
18.	$\forall x\, (P(x) \to T(x))$	$\to$e, 1, 9
19.	$\forall x\, (T(x) \to H(x))$	$\to$e, 2, 15
20.	y_0	fresh variable
21.	$P(y_0) \to T(y_0)$	$\forall$e, 16
22.	$T(y_0) \to H(y_0)$	$\forall$e, 17
23.	$P(y_0)$	assumption
24.	$T(y_0)$	$\to$e, 19, 21
25.	$H(y_0)$	$\to$e, 20, 22
26.	$P(y_0) \to H(y_0)$	$\to$i, 21–23
27.	$\forall x\, (P(x) \to H(x))$	$\forall$i, 18–24
28.	$\exists x\, (T(x) \wedge H(x)) \to \forall x\, (P(x) \to H(x))$	$\to$i, 3–25